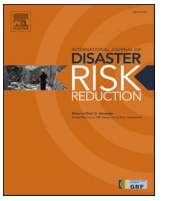
International Journal of Disaster Risk Reduction 127 (2025) 105636



Contents lists available at ScienceDirect

International Journal of Disaster Risk Reduction

journal homepage: www.elsevier.com/locate/ijdrr





A novel clay shrink-swell buildings damage model: From unstructured insurance data to the creation of buildings database, and the proposition of damage severity scale

Noel Philmon Mzungu [a,b,c,*], Caterina Negulescu [b], Etienne Bertrand [d], Arthur Perringaux [a], Laurent Boissier [a,c], Sébastien Gourdier [b], Bastien Colas [e], Frédéric Leone [c], Gilles Grandjean [b]

[a] Generali France (Generali Climate Lab), 2 rue Pillet Will, 75009, Paris, France
[b] BRGM, French Geological Survey, Orléans, France
[c] Univ Paul Valéry Montpellier, LAGAM, F34000, Montpellier, France
[d] Université Gustave Eiffel, IFSTTAR GERS/SRO, 14-20 Boulevard Newton, Champs-sur-Marne, 77447, Marne-la-Vallée, France
[e] BRGM, French Geological Survey, Montpellier, France

A B S T R A C T

Expansive clay soils undergo seasonal moisture fluctuations, swelling when wet and shrinking during dry periods. These volumetric changes induce differential ground movements beneath buildings, often resulting in structural damage, particularly in low-rise residential buildings. In France, this phenomenon has become increasingly costly, with average annual insurance claims of 1 billion euros between 2016 and 2021, rising to 3.5 billion euros in 2022 due to severe droughts and heatwaves driven by climate change.

Despite its growing impact, damage assessment remains challenging due to the scarcity of empirical data and the localized variability of soil-structure interactions. To address this, we propose a scalable, data-driven methodology that combines expert knowledge and advanced natural language processing to enable empirical risk analysis.

This study introduces the Clay Shrink-Swell Damage Severity Scale, constructs a structured database of building and environmental characteristics, and identifies damageability factors based on observed damage levels. A total of 10,325 loss adjustment reports from Generali (2000–2021) were analysed. A subset of 155 reports was manually annotated and scaled to the full dataset using Large Language Models (LLMs) under a Retrieval-Augmented Generation (RAG) framework.

The resulting tools provide a solid foundation for post-event assessments, support rapid field diagnostics, and enable the development of personalized prevention strategies. This work contributes to improved vulnerability modelling and more effective risk reduction in a changing climate.

1. Introduction

Expansive clay soils respond strongly to seasonal variations in moisture content, shrinking during dry periods and swelling when rehydrated [1,2]. These volume changes can lead to differential ground movement, which is a frequent cause of structural damage, particularly in lightweight residential buildings. Cracks in walls, distorted openings, and foundation shifts are typical consequences [3–9]. These damages raise concerns for homeowners, insurers, and local governments.

Several countries, including France, the United Kingdom, the United States, Australia, and South Africa, have reported substantial

* Corresponding author. Generali France (Generali Climate Lab), 2 rue Pillet Will, 75009, Paris, France.
E-mail address: noelphilmon.mzungu@generali.com (N.P. Mzungu).

https://doi.org/10.1016/j.ijdrr.2025.105636
Received 7 February 2025; Received in revised form 6 June 2025; Accepted 6 June 2025
Available online 10 June 2025
2212-4209/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

building damages linked to this phenomenon [8–11]. In France, the focus of this study, shrink-swell activity peaks during summer, when clay soils dry after winter saturation [12]. This phenomenon is often referred to as "geotechnical drought risk," highlighting its seasonal dynamics and structural impacts. Following the 1976 drought, in 1989, this risk was officially recognized by the national natural catastrophe insurance system (Régime Cat-Nat) [13,14]. Its associated damage costs have continued to grow, reaching 3.5 billion euros in 2022 [15].

However, despite this rising financial burden, shrink-swell related damage remains underrepresented in natural hazards and climate risk research [14]. Various studies have approached the clay shrink-swell phenomenon using different data sources and methods. While previous studies have primarily focused on the geological and geotechnical characteristics of clay soils, examining their distribution, behaviours, and links to damage costs [6,16], building vulnerability itself has received comparatively little attention. Yet, understanding the factors that influence structural fragility in the context of drought-induced subsidence is essential. Key questions arise: Should we reinforce foundations, stabilize the soil, implement environmental mitigation strategies, or adopt a combination of these approaches to enhance building resilience?

Some have relied on meteorological models and climate data [14,17,18], while others have used insurance records to statistically estimate drought frequency and intensity [19]. Damage data have also been analysed to explore building vulnerability [6,51,52,54,55]. However, vulnerability assessments based on basic geometric ratios, microanalysis, or wall mechanics remain limited. More advanced methods, such as evaluating bracing systems, justifying damage mechanisms, and designing reinforcement strategies for probabilistic risk assessments, are valuable but often too complex for the insurance sector, which mainly deals with masonry and single-family homes [20]. Some civil engineering studies have shifted from simplified beam models to soil-structure interaction approaches, though these can misrepresent actual damage levels and are not well suited for large-scale risk modelling [21,49,50].

Ultimately, addressing the impacts of clay shrink-swell induced soil movements on buildings remains difficult when engineering, geotechnical, climatic, and statistical perspectives are considered in isolation. There is a notable gap in research addressing the physical vulnerability of buildings to clay shrink-swell, especially at regional scales. Most vulnerability studies focus on other hazards such as floods, landslides, and earthquakes, with few dedicated to this specific phenomenon [22–28]. Building damageability can be assessed using empirical or analytical/mechanical approaches. The empirical method relies on post-event data to link hazard characteristics with observed damage. It is relatively fast and grounded in real-world evidence but requires extensive data [21]. The analytical approach, on the other hand, uses simulations to explore structural responses across different building types.

Method selection depends on the scale and goals of the study [26]. For large-scale assessments, such as at the national level, the empirical approach is generally favoured due to its practicality in handling extensive datasets. Accordingly, this study adopts the empirical method to evaluate building damageability. Moreover, there is a lack of public, standardized post-event assessments that document the relationship between building characteristics and observed damage patterns. As noted by Douglas [29] and Negulescu et al. [21], unlike disasters such as earthquakes, where post-event damage surveys are common and often publicly available, detailed assessments of damage from clay shrink-swell are limited. This is partly because such events rarely result in human casualties, reducing the perceived urgency to develop dedicated vulnerability models.

The scarcity of historical damage data presents a significant barrier to developing reliable models for assessing building vulnerability [30]. In this context, insurance records, particularly loss adjustment reports, offer a valuable alternative source of empirical information. In France, insurance companies systematically conduct post-event assessments for claims related to clay shrink-swell phenomenon [15]. Certified loss adjusters apply a standardized methodology that combines preliminary desk-based analysis with on-site inspections.

The initial phase involves reviewing geological and geotechnical maps, groundwater datasets, and national hazard databases (including the BRGM clay shrink-swell susceptibility map) to assess the likelihood of shrink-swell processes. This step also supports the exclusion of alternative causes of damage, such as mining subsidence or karst dissolution. During the subsequent site visit, adjusters evaluate the condition of the building, document crack patterns and deformation, assess surrounding vegetation, and collect structural information when accessible, particularly regarding foundation type and layout. Although root depth is not directly measured, risk is inferred from the ratio of tree height to its distance from the building. Intrusive geotechnical investigations, such as soil sampling or trial pits, are not included in routine procedures due to cost and scalability constraints, though they may be recommended for complex or disputed cases.

Each loss adjustment report concludes with a comprehensive synthesis, including damage classification, cause attribution, cost estimation, and expert recommendations. A final section explicitly justifies whether the observed damage is attributed to clay shrink-swell activity or to another cause, based on the available evidence. These reports are essential tools for insurers, informing compensation decisions (especially in relation to CatNat declarations), guiding risk management strategies, and supporting regulatory compliance.

While the level of detail varies across reports, particularly for foundation depth and subsurface root presence, the expertise process remains a cornerstone for understanding real-world building vulnerability. This study builds on that foundation to model damageability using the best available indicators and highlights the need for future research that integrates more detailed geotechnical data. Loss adjustment reports, like other insurance data, are confidential due to client privacy and commercial considerations, which limits their use in risk assessment studies [31,32]. Nonetheless, such data have been used in local [32,33] and international studies [51,53] to analyse natural hazards, including clay shrink-swell phenomenon. This study encountered two main challenges when working with these unstructured reports. The first was transforming descriptive text into a structured database using specific keywords to identify structural systems and vulnerabilities. The second was standardizing damage descriptions into a consistent damage severity scale. Both steps are essential for generating accurate clay shrink-swell risk scenarios, yet no such structured database currently exists.

To extract information from unstructured text, researchers use methods such as traditional text mining, natural language

processing (NLP), and large language models (LLMs). Text mining relies on identifying predefined word sequences or key phrases. It is easy to implement and effective for targeted extraction but lacks contextual awareness and may overlook relevant content not captured by selected keywords. Manual methods and basic text mining have proven inadequate when processing large volumes of reports [33–36]. To overcome these limitations, NLP techniques offer more advanced analysis by considering syntax, structure, and semantics. However, despite being more robust than simple keyword searches, NLP may still struggle with deeper context and ambiguity [33,35,36].

Since 2022, large language models (LLMs) such as GPT-3 and GPT-4 have shown significant promise in text analysis and information extraction. These advanced models can interpret complex instructions and extract contextually relevant information, even when it does not match predefined keywords. LLMs excel at recognizing linguistic nuances, learning complex patterns, and identifying semantic relationships. They can also process structured data, including tables, charts, and diagrams. When guided by human input, LLMs offer an effective solution for handling the complexity and scale of data required for empirical risk assessments [37–39]. However, LLMs are not without limitations. They may produce inaccurate outputs, particularly when clear information is lacking, and can suffer from issues such as hallucination, outdated knowledge, and opaque reasoning. To mitigate these drawbacks, Retrieval-Augmented Generation (RAG) has emerged as a complementary approach, enabling models to incorporate up-to-date, verifiable information from external databases[40].

This paper presents the use of unstructured insurance damage data to propose the Clay Shrink-Swell Damage Severity Scale. It outlines the construction of a detailed database combining building features and environmental context. The study identifies damageability factors by linking observed damage grades to specific building and site characteristics. However, connecting damage grades directly to physical hazard processes was beyond the scope of this study. The analysis presented here is exploratory and limited to correlations, without asserting causation. The relationship between correlation and causation will be addressed in a forthcoming study.

2. Methodology

This study aims to identify factors contributing to building damageability to clay shrink-swell by analysing correlations between observed damage grades and the characteristics of buildings and their surroundings, based on information extracted from insurance loss adjustment reports. The methodology is organized into three main stages (Fig. 1):

- **Database development**
- **Proposition and application of a damage severity scale specific to clay shrink-swell**
- **Identification of damageability factors through statistical correlation analysis**

Section 3 presents the database development process, including data collection (3.1), manual creation of a reference dataset (3.2), and the application of a large language model (LLM) within a Retrieval-Augmented Generation (RAG) framework (3.3). Section 4 outlines the proposition of the clay shrink-swell damage severity scale (4.1), the standardization of damage criteria (4.2), and the attribution of damage grades to the dataset (4.3). Finally, we describe the statistical analysis conducted to identify damageability factors based on correlations between damage grades and building or environmental features.

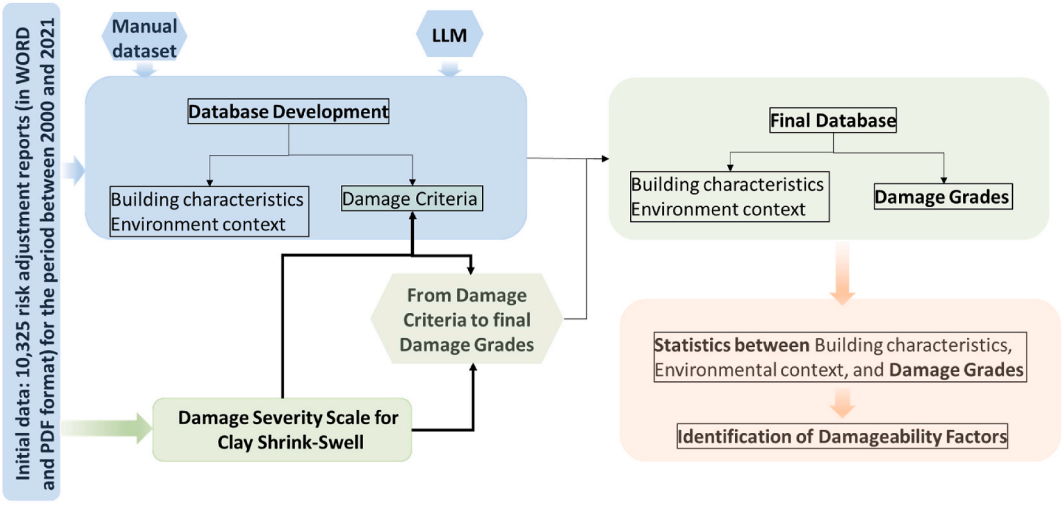


Fig. 1. Methodology used in this study. Blue represents the development of the database; green corresponds to the proposition of the damage scale and classification of damage grades in the final database; orange indicates the statistical analysis and identification of damageability factors.

3. Database development

A detailed database of buildings characteristics and environmental context was developed to analyse damageability to clay shrink-swell, with other causes of subsidence (e.g., mining or rock dissolution) explicitly excluded. The database was built using insurance loss adjustment reports, following a three-step process (Fig. 2): (1) manual creation of a reference dataset from 155 reports, (2) prompt engineering and calibration of a large language model (LLM), and (3) automated extraction from 10,325 reports using the calibrated model.

3.1. Data collection

This study uses loss adjustment reports provided by Generali Insurance Company, covering the period from 2000 to 2021. These unstructured reports (in Word and PDF formats) follow a standardized template introduced in 1999 by the French Federation of Insurance Companies (FFSA) and the Group of Mutual Insurance Companies (GEMA), specifically for assessing damage related to clay shrink-swell [33]. The selected time frame reflects the availability of consistent reporting practices and the typical lag between hazard occurrence and damage manifestation.

The dataset offers broad geographic coverage of mainland France (Fig. 3). Reported damages are concentrated in areas with high exposure, particularly in the southwest, and align closely with national exposure maps produced by the French Geological Survey [16].

The information contained in these reports can be categorized into three main groups:

1. **Administrative information**: Includes details about the insured, such as names, contacts, address, ownership status, etc.
2. **Technical assessment operations**: this section is central to the damageability analysis conducted in this study. It includes **building characteristics**, **environmental context**, and **damage observations** (Table 2)
3. **Expert opinion:** This stage involves the loss adjuster's assessment of whether the damage is attributable to clay shrink-swell phenomena. Where uncertainty remains, further geotechnical investigations may be recommended (Table 1).

3.2. Manual creation of a reference dataset

Fields were grouped into two main categories: building characteristics and environmental context (Table 2). These fields were clearly identifiable, as the loss adjustment reports followed a standardized structure (Section 3.1). Our task was to validate their relevance based on existing literature [41] and ensure they were consistently represented. We also refined keyword definitions and classification schemes through iterative feedback to address data heterogeneity and improve consistency.

Several automated methods, including LLMs and keyword-based extraction, were tested against this reference dataset. The LLM-based approach yielded the highest accuracy (see Section 3.3) and was subsequently applied to the remaining reports.

The reference dataset also played a key role in prompt engineering. It enabled iterative fine-tuning of the LLM prompts, enhancing precision and reliability in data extraction.

3.3. Application of large language model (LLM) under the Retrieval-Augmented Generation (RAG)

To scale up the manually created reference dataset from Section 3.2, we applied the Large Language Model (LLM) within the RAG framework. This approach enabled the extraction and classification of information from unstructured insurance reports, offering

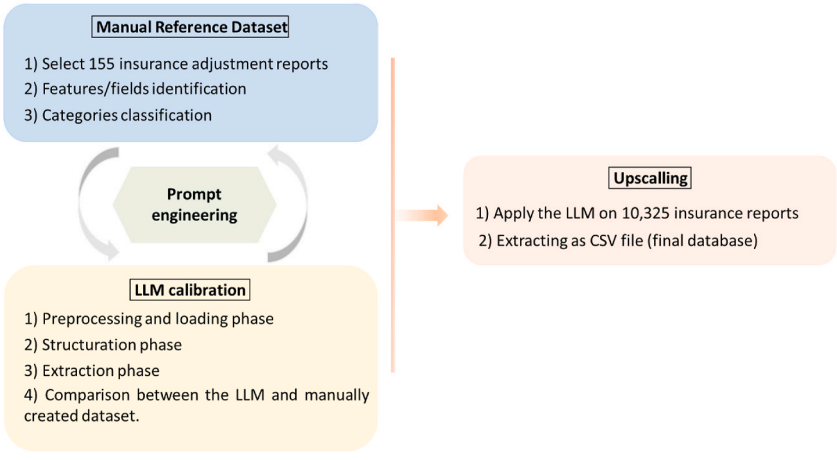


Fig. 2. Iterative process for extracting information from loss adjustment reports: manual creation of a reference dataset, prompt engineering, and application of the LLM to process 10,325 reports and generate final database.

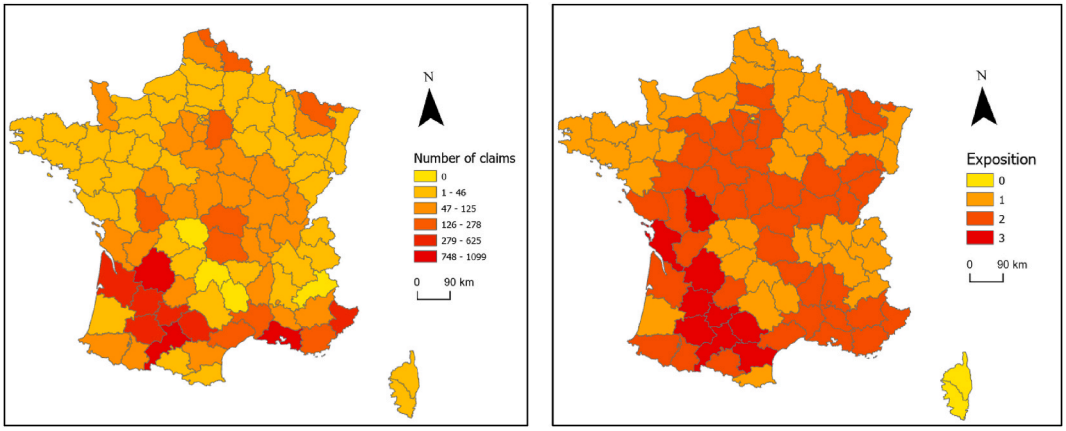


Fig. 3. Correlation maps of mainland France by departments. Left: Number of reported damage claims. Right: Clay shrink-swell exposure levels [16].

Table 1
Categories assigned by loss adjusters for recommended building repair following damage from clay shrink-swell phenomenon.

Category	Expert recommendations'
0	Soil study is recommended
1	Disorders not attributable to drought (clay shrinkage and swelling)
2	Crack treatment
3	Need to repair foundations and/or superstructures
4	Need to create special foundations or implement other means of soil stabilization
5	Total loss (building economically irreparable)

Table 2
Structure of the database: Main blocs and specific fields describing building characteristics and surrounding environmental conditions.

Main blocs	Specific database field required
Building characteristics	Location
	Type (detached or nondetached)
	Year of construction
	Dimensions
	Extensions and modifications
	Substructures (presence of basement, etc.)
	Foundations (type and depth)
	Superstructure (floor, roof, internal and external walls, etc.)
Environmental context	Topography of the terrain
	Soil index properties
	Soil type
	Vegetation (distance and height)
	Recent infrastructure works around the building
	Watercourses
	Aquifers, etc.

improved accuracy and contextual understanding compared to traditional methods [39,42,43]. RAG was selected for its ability to guide LLMs in retrieving and interpreting relevant information from external sources when needed [40].

We treated our data with the LLM through three phases:

- **Preprocessing and loading phase**

This phase involved the document parsing and vectorization processes. We first parsed all the documents using the Microsoft azure optical character recognition (azure OCR) and Custom Block. We then vectorised the documents using the open AI embedding model (text-embedding-ada-002) to get vectors of 1572 dimension. So, to embed all the documents, the pages were considered as chunks i.e., 1 page = 1 chunk (chunking refers to a process of breaking large piece of text into smaller segments). Thereafter, everything was put inside the vector store. In our case we used Chroma DB vector store which is an open-source vector database for AI.

- **Structuration phase**

This is the retrieval step where for each document we designed a prompt (instructions) as an input that reflects our reference dataset. We then performed the retrieval stage to get the most relevant chunk according to the prompt.

- **Extraction phase**

In this phase all the above retrieved chunks were passed as a context to the LLM GPT4 to extract the relevant information.

To ensure consistency in data extraction and classification, we compared the LLM output with the manually created dataset. The comparison focused on two key aspects: extraction accuracy and classification consistency. For extraction, we evaluated whether the LLM successfully retrieved the same information captured manually. In 7 cases out of 155, the LLM failed to identify data that had been manually recorded, resulting in a 95 % extraction accuracy. For classification, we assessed whether the LLM assigned the same categories as in the manual dataset. In 30 cases, the LLM proposed different classifications, often identifying additional modalities that had not been captured manually. This resulted in an 80 % classification accuracy. We refined the prompt iteratively to improve model performance until satisfactory accuracy was achieved.

4. Proposition of the clay shrink-swell damage severity scale, damage criteria and attribution of damage grades

In parallel with the database development, a detailed review of the loss adjustments reports enabled the proposition of a dedicated damage severity scale for clay shrink-swell, based on observed structural impacts recorded in historical reports.

4.1. Damage severity scale

To identify building damageability factors, it was essential to link damage grades to building and environmental characteristics. We developed the scale by analysing descriptions, photographs, and sketches from the same 155 reports as in Section 3.2. The scale draws on established frameworks, notably the Burland (1975) scale, which classifies damage based on crack width, and the European Macroseismic Scale – 1998 [44], which distinguishes structural from non-structural damage. Loss adjustment reports were categorized into five levels of severity, from D1 (minor damage) to D5 (severe damage), based on qualitative comparison with the two reference scales and the specific characteristics of clay shrink-swell.

To validate the classification, we organised a workshop with two independent groups of experts, consisting of both insurance professionals and researchers. Each group reviewed the same 155 reports, focusing only on photos and sketches, and assigned damage levels based on their expertise. While some discrepancies occurred (e.g., level 2 vs. level 3), there was strong overall agreement. This expert consensus informed the final version of the scale, in Fig. 4.

Our damage scale is largely based on the Burland scale (1975) but is distinguished by its foundation in real-world cases, using empirical damage data from insurance reports. This gives it greater practical relevance compared to scales derived primarily from theoretical modelling. By harmonizing language across reports, the scale is accessible to both professionals and non-specialists.

A key feature of our scale is its differentiation among exterior damage, building damage, and foundation problems, allowing for

Burland (1975)	Level of Damage CSS-Dx	Damage descriptions			Damage Keywords	Crack Width Thresholds	Description of Crack Widths	Crack Location	Crack Evolution
		Building	*Foundations*	*Exterieurs*					
D0 (negligible) and D1 (very slight): Superficial cracks mainly caused by shrinkage and expansion effects & Fine cracks in brick or masonry walls	D1	Microcracks, some tile cracks	Not applicable	Cracks in the terrace, cracks in the walls between blocks (*e.g.*, dovecote and garage), cracks in the retaining wall, fence cracks, subsidence of the exterior terrace at the junction with the main structure of the house.	Vertical hairline crack on the wall between the terrace and garden, crazing of the render, sidewalk subsidence, vertical microcracks, multidirectional cracks in the basement slab, masonry joint cracks, horizontal crack in the pier, horizontal gap between the paving and entry step, subsidence of the external staircase paving, tile joint.	0,1 mm to 1 mm	One or several cracks, maximum 1mm in size	structural elements (lintels, piers), wall connections, tile joints, coatings, hard floor coverings (tiles, parquet), facades (crazing), junctions in plasterboard walls and ceilings	No
D2 (slight): Visible cracks, and windows and doors may slightly jam	D2	Cracks at door and window levels, cracks in non-structural elements, cracks under structural elements, tile cracks	Not applicable	Junction of two structures, cracks in boundary walls, cracks in the base wall.	Horizontal cracks extending to patio door openings, oblique cracks on both sides of the window, oblique crack on the partition wall, horizontal crack on the tiles, cracking at the junction of two structures, crack on the boundary walls.	1 mm to 5 mm	A crack between 1 and 5mm wide, or several cracks less than 1mm wide	**House:** doors, windows, windows, partitions, tiling, wall junctions **Exterior & Environment:** fence, foundation wall	No
D3 (moderate): Clearly visible cracks, pipelines and service pipes may be broken	D3	Clearly visible cracks in the house, non-through cracks, cracks between structural and non-structural elements (which outline lintels and bricks)	Not applicable	Significant cracks in the environment, sidewalks, terraces, and pipelines, as well as potential breakage of service pipes.	Horizontal cracks caused by floor rotation, horizontal crack outlining the lintel, joint cracks, 45° crack outlining the masonry bond, cracks in the wall lining.	5 mm to 15 mm	Either one crack in the range of 5 mm < L < 15 mm or multiple cracks ranging from 1 to 5 mm	Multiple locations in structural elements: walls, gables, facades (at least 2-3)	Yes
D4 (severe): Generalized cracks, especially at the corners of openings (mustache cracks), slight wall inclination (loss of plumb, bulging walls), twisted doors and windows, sloping floor, loss of bearing capacity of certain beams, fractures in structural elements, number of cracks and spacing between cracks to be considered	D4	Widespread cracks, especially in the opening corners (hairline cracks), multiple walls with cracks larger than 10mm, twisted doors (difficulty opening and closing the sliding glass door), multiple fissures in the flooring	Foundation crack, settlement/settlements/ undermining of the base	Not applicable	Horizontal crack with a maximum offset thickness greater than 3 mm: 15 mm, old settlement of 20 cm, open cracks in structural elements and masonry, significant undermining of the footing of the partition wall, cracks opening between 10 to 15 mm, structural horizontal cracks due to floor rotation, slab cracks.	15 mm to 25 mm	Either one crack in the range of 15 mm < L < 25 mm or a minimum of three cracks larger than 10 mm	Multiple locations in structural elements: foundation, load-bearing walls, framework, floor slabs, gables, facades (at least 2-3)	Yes
D5 (very severe): Extensive cracks, tilting walls (leaning dangerously), broken and warped windows, instabilities, loss of bearing capacity of certain beams, fractures in structural elements, numerous cracks to be addressed	D5	Widespread and extensive cracks, multiple walls with cracks larger than 10mm or one crack larger than 25mm, twisted doors (difficulty opening and closing the sliding glass door), multiple cracks in the flooring	Foundation crack, loss of bearing capacity, settlement/settlements/ undermining of the base, instability	Not applicable	Very wide cracks, through cracks, partial collapse, cracking of almost all ceiling corners, terrace subsidence, subsidence > 1 cm, wall separation, wall deformation/bending, crack at the top of the foundation, offset at the corner, crack across the entire facade, crack at the junction between foundations and concrete walls, generalized floor subsidence, misaligned stairwell door, vertical crack extending to the foundations, slab cracks.	Above 25 mm	Several (at least 3-4) different locations in the structural elements and at the foundation level	Several (at least 3-4) different locations in the structural elements and at the foundation level	Not applicable

Fig. 4. The proposed damage severity scale specific for clay shrink-swell phenomenon from the empirical buildings damage data.

more accurate diagnosis. Exterior damage, such as external cracks, often signals emerging structural issues and highlights the need for early intervention. Building damage confirms actual deterioration of components and requires distinguishing between structural and non-structural issues. Structural damage compromises a building's integrity and is costly to repair, while non-structural damages are superficial and less critical. Foundation problems are particularly urgent, as they affect the building's stability and often require specialized interventions like underpinning or soil stabilization. Early detection is essential to prevent widespread damage and ensure safety. By distinguishing these types of damage, the scale supports timely and appropriate responses, helping to preserve structural integrity and reduce repair costs.

Therefore, the proposed damage scale contains the following fields:

1. **Damage levels**: Like many other scales, our scale has 5 levels of damage ranging from D1 to D5 [45]. D1 represents minor damage such as hairline crack, while D5 corresponds to severe cases like wall collapse and settlement.
2. **Description of damage**: Damages are categorized based on where they appear:
   - **Exteriors**: Minor damages affecting external elements such as sidewalks or fences, with minimal impact on the main structure.
   - **Building**: Damages within the house, affecting non-structural elements, both structural and non-structural elements, or structural elements alone.
   - **Foundations**: Significant structural damages affecting building's foundations, often indicating serious structural compromise.
3. **Keywords**: A curated list of terms describing crack patterns, supporting accurate classification.
4. **Crack width intervals (D1 to D5)**: This criterion remains a key classification parameter, though it should not be used alone.
5. **Crack width description**: To capture the severity more accurately, this field includes qualitative details. A single large crack may be as significant as multiple smaller ones, which justifies the inclusion of this field.
6. **Crack evolution**: This field is highly important as it tracks whether cracks are stable or worsening. An evolving crack widening, deepening, or spreading, may indicate ongoing structural deterioration and increased damageability.

4.2. Damage criteria

To address this, we developed our own structured damage criteria based on the proposed damage scale (see Section 4.1). The iterative process linking the scale and criteria is illustrated in Fig. 1. In contrast to building and environmental characteristics, where fields were predefined in the insurance reports, damage characterization posed a greater challenge. Experts (risk adjusters) used free-form language, without standardized formatting or keywords. To address this, we developed our own structured damage criteria based on the proposed damage scale (see Section 4.1). The iterative process linking the scale and criteria is illustrated in Fig. 1.

After establishing the five-level scale (D1 to D5), we extracted most common damage descriptors associated with each level. Given the variability and occasional incompleteness of the reports, we focused on four key variables: crack width (W), crack location (L), crack shape (S), and crack kinetics (K). Although other variables (e.g., number or length of cracks) could be informative, they were often missing and thus excluded from our analysis. Thus, the list of damage variables used is not exhaustive but adequate for describing the observed damage in this study.

Crack width (W): This is a critical but not standalone indicator of damage severity [46,47]. While wider cracks generally indicate more severe damage, width must be interpreted alongside other factors. We adopted the Burland (1975) scale for crack width intervals, as these closely aligned with the measurements found in our reports (Fig. 5).

Crack kinetics (K): This variable refers to the mode of crack movement and is essential for understanding the cause, severity, and appropriate repair strategy [48].

Depending on how the foundation moves relative to the crack plane, crack faces may shift horizontally, vertically, or slide along each other. Using our proposed damage scale, we characterized and ranked different crack propagation patterns to better understand the evolution of damage (Fig. 6).

Some cracks exhibited **no displacement**, often indicating minor damage. **Infiltrative cracks**, which allow air and water to enter without necessarily penetrating the full depth, may gradually weaken the structure if left unaddressed. Clay soils' drying and wetting cycles can cause **cracks to open and close repeatedly**, increasing water infiltration during rainfall and exacerbating soil swelling. This cyclic behaviour contributes to long-term deterioration of foundations.



Fig. 5. Examples of cracks observed in different parts of the building, illustrating various cracks openings and structural contexts: (a) Crack on the ceilings of the extension; (b) Subvertical crack with an opening; (c) Separation cracks between the concrete slab of the terrace and the base of the façade; (d) Fine hairline cracks extending from the lintels of a bedroom window.

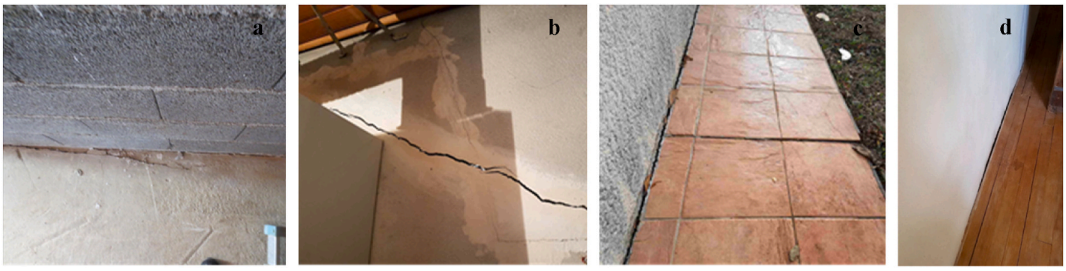


Fig. 6. Examples of different crack opening modes observed in residential structures: (a) Settlement of the concrete slab at the base of the wall; (b) Through-wall crack above ground level; (c) Peripheral sidewalk settlement leading to tile detachment; (d) Gap formation between the interior bedroom wall and the wooden floor.

Crack propagation involving vertical or horizontal displacement often signals more severe structural impacts. **Settlement** represents the most advanced stage of damage, where part of the building sinks due to uneven ground movement. In such cases, issues like jammed doors and windows, ruptured pipes, and visible foundation distress may indicate that the structure has become unsafe.

Crack shapes (S): Crack morphology conveys critical information about both the severity of structural damage and its underlying causes. This study focused on three principal crack types: vertical, stair-step (zigzag), and horizontal cracks (Fig. 7). The implications of crack shape depend on various contextual factors, including location, extent, and associated damage mechanisms.

- **Vertical cracks** run perpendicular to the ground, typically appearing in an up-and-down orientation. Their structural impact is often assessed based on their width and location. While generally considered less severe than other crack types, they should not be overlooked, as they may signal deeper structural issues that could evolve over time.
- **Stair-step (zigzag) cracks** follow mortar joints in masonry walls, forming a stepped or angular pattern. These are commonly associated with foundation movement and differential settlement, both of which are frequent outcomes of shrink-swell soil behaviour. Their presence is a strong indicator of structural stress at the ground level.
- **Horizontal cracks** extend parallel to the ground and are particularly concerning when found on foundation walls or substructures. They can indicate lateral soil pressure or, in clay-rich soils, foundation upheaval due to swelling. Such cracks pose a significant threat to structural integrity and typically warrant immediate investigation and remediation.

By categorizing and interpreting crack shapes, we aim to better link observed damage to causal processes and assess vulnerability with greater precision.

Crack location (L): As outlined in the damage severity scale, the location of cracks within a structure plays a critical role in assessing the severity and implications of the damage. Cracks may appear on external surfaces, internal walls, or structural elements, and each location provides distinct insights into potential failure mechanisms and building vulnerability.

Cracks affecting structural elements, such as load-bearing walls, beams, columns, and foundations are more critical than those found in non-structural components like plaster or cladding. This is because damage to structural elements can directly compromise the building's stability and load distribution capacity.

Among all locations, foundation cracks are typically the most serious (Fig. 8). The foundation is responsible for supporting the entire structure and transmitting loads to the ground. Cracking in this zone may indicate ground movement, settlement, or heave associated with shrink-swell activity, and often poses a significant risk to overall structural integrity.

Accurate identification and interpretation of crack location is thus essential for determining both the cause and potential consequences of damage.



Fig. 7. Representation of different shapes of cracks as they could be seen in different parts of the building: (a) A significant vertical crack marks the physical separation between the garage and the neighbouring property (b) Step crack on the retaining wall at the garage's corner, (c) Horizontal crack extending along the length of the garage adjacent to the old neighbouring building.



Fig. 8. (a) Cracking of the previously filled concrete slab of the terrace (b) Cracks around the perimeter of the left window (c) Cracks on the ceiling (d) The crack on a fence that is wider at the top than at the bottom.

4.3. From damage criteria to Final Damage Grades in the database

Damage grade translates commonly observed damage patterns from ordinal scale categories into numerical values, with higher numbers indicating more severe damage [30]. To integrate the proposed damage criteria into the database, we translated descriptive information into standardized damage levels (D1 to D5) across the four key variables. Drawing inspiration from the French Association for Earthquake Engineering (AFPS) emergency diagnosis methodology (Taillefer et al., 2019), we adopted a holistic approach. We combined various damage characteristics (e.g., crack width, location) affecting different parts of the building (environment, structure, parking lot) to determine a degree of damage, considering all observations collectively. These variables were then categorized into five damage levels, providing a structured framework for subsequent analysis, as summarized below (Figs. 9–12).

Crack width remains a widely used reference for evaluating structural damage. The analysis of crack widths (Fig. 9) showed that 42 % of observed cases had widths between 1 and 5 mm, 31 % ranged from 0.1 to 1 mm, 22 % fell within the 5–15 mm range, and 5 % exceeded 15 mm. These measurements provide critical insight into damage severity and support the broader classification of building damageability.

Crack kinetics: An analysis of crack kinetics (Fig. 10) revealed that 55 % of the cases involved penetrating cracks with displacement, indicating significant structural movement. 27 % of cases exhibited cracks without displacement, suggesting more superficial or early-stage damage. 10 % showed evidence of settlement-related deformation, while 7 % involved cyclical opening and closing, typically associated with seasonal shrink-swell cycles. These findings offer valuable insight into the dynamic behaviour of damage over time and help differentiate between reversible and progressive structural responses.

Crack location: Analysis of crack location data revealed distinct patterns across structural and non-structural components (Fig. 11). A total of 41 % of the cases reported damage at the foundation or infrastructure level, highlighting the critical role of ground-structure interaction in shrink-swell processes. 32 % of cracks were observed in various non-structural elements, including coatings, crazing, partition joints, drywall, exterior surroundings, terraces, porches, sidewalks, and ceilings. 14 % of cases involved hard flooring, doors, and windows, while 8 % were associated with structural joints, façades, and lintels. The remaining 5 % affected gable walls, floor slabs, and load-bearing walls. These distributions reflect the variability of crack occurrence across building components and reinforce the importance of spatial context in damage assessment.

The analysis of **crack shapes** revealed a variety of configurations reflecting different stress patterns and structural responses (Fig. 12). A total of 29 % of cases exhibited a combination of horizontal and stair-step or horizontal and vertical cracks. 27 % presented a mix of horizontal, vertical, and stair-step cracks, indicating complex structural deformation. 22 % showed either horizontal cracks alone or a combination of vertical and stair-step cracks. 17 % involved stair-step cracks only, typically associated with differential settlement, while 5 % displayed vertical cracks alone, often considered less severe but still requiring monitoring. These distributions help characterize the mechanisms driving observed damage.

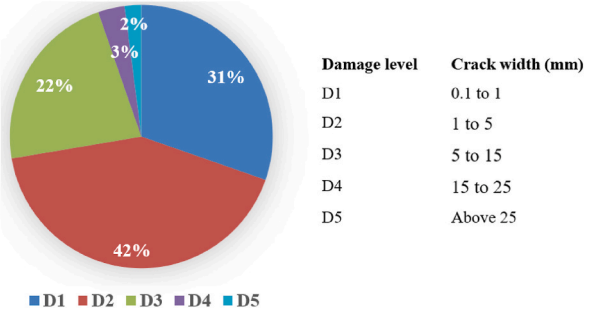


Fig. 9. Distribution of crack width (in mm) based on available data. The analysis includes 4397 cases out of 10,325 total records, representing 43 % of the dataset with reported crack width information.

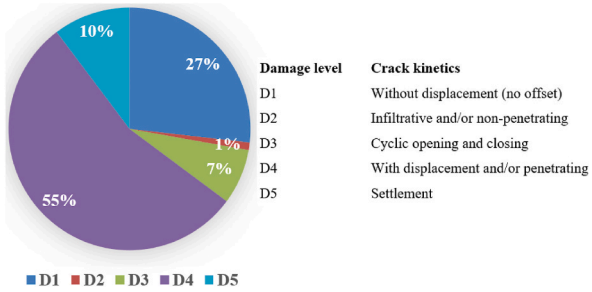


Fig. 10. Distribution of crack opening modes based on available data. The analysis includes 2187 cases out of 10,325 total records, representing 21 % of the dataset.

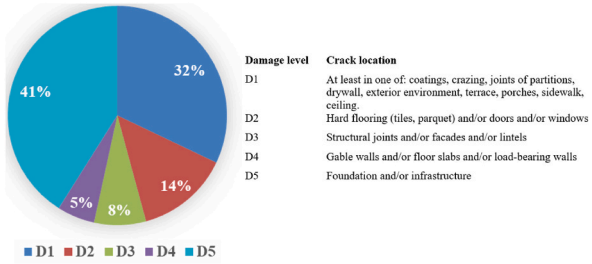


Fig. 11. The distribution of crack on different locations outside and within the building itself, based on the cases with available information, 7403 cases out of 10,325 (representing 72 %).

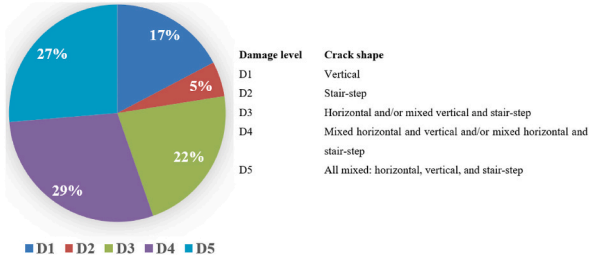


Fig. 12. Distribution of crack shapes or patterns based on available data. The analysis includes 6351 cases out of 10,325 total records, representing 62 % of the dataset.

Table 3
Distribution of damage criteria fields in the database, illustrating the challenge of synthesizing multiple attributes into a final damage grade across the four criteria.

Crack width (W)	Crack kinetics (K)	Crack shapes (S)	Crack location (L)	Count	%
–	–	–	–	2696	25,99
W	–	S	L	2431	23,44
–	–	S	L	1793	17,29
W	K	S	L	1519	14,65
–	–	–	L	789	7,61
–	K	S	L	453	4,37
W	–	–	L	250	2,41
–	–	–	S	95	0,92
W	K	–	L	93	0,90
–	K	–	L	75	0,72
W	–	–	–	44	0,42
W	–	S	–	40	0,39
–	K	–	–	22	0,21
W	K	S	–	15	0,14
–	K	S	–	5	0,05
W	K	–	–	5	0,05

4.4. Final database with damage grade

Our final database comprises 10,325 records, each enriched with damage grades derived from four key criteria: crack width, crack kinetics, crack shape, and crack location, as previously described.

A major methodological challenge involved synthesizing these individual damage dimensions into a single, final damage grade. For instance, a given case might be assigned a grade of D4 for crack width, D3 for crack kinetics, D1 for crack shape, and D2 for crack location. This raises the question: How should these be combined into one representative damage grade?

Furthermore, not all records contain complete information for all four criteria (Table 3). In some cases, damage grades were available for only one, two, or three criteria, while others had none.

It is important to emphasize that there is no established formula for aggregating the four damage criteria into a single damage grade. We initially explored several approaches, including machine learning techniques such as random forest and linear discriminant analysis (LDA), but these methods yielded unsatisfactory and unstable results. Consequently, we adopted a simple linear combination, assuming that all four criteria contribute equally to the overall damage assessment.

For records with complete information on all four criteria, we computed the arithmetic mean of the corresponding damage grades. To test the robustness of this method, we conducted simulations by randomly suppressing one or two criteria and recalculating the average. Results showed that this approach remained consistent and reliable, even in the presence of missing values.

Based on this strategy, we compiled a final dataset containing the computed damage grades, along with detailed information on building characteristics and environmental context.

5. Results

This section presents the results of our empirical assessment of building damageability to clay shrink-swell phenomenon, based on a comprehensive statistical analysis of observational damage data. The dataset was compiled from loss adjustment reports provided by Generali Insurance Company, covering the period from 2000 to 2021.

The final database consists of 10,325 records and 98 variables, organized into three main categories:

- Building characteristics (e.g., year of construction, foundation type, structural materials),
- Environmental contexts (e.g., terrain topography, vegetation presence), and
- Damage grades, ranging from D1 (minor damage) to D5 (severe damage).

While the analyses presented are preliminary, they provide meaningful and actionable insights, demonstrating the potential of the tools developed in this study. Emphasis is placed on the **damage severity scale**, which serves as a key innovation for standardizing and comparing damageability factors. This scale enhances the interpretability and applicability of clay shrink-swell damage data and supports the development of more robust and spatially explicit risk assessments.

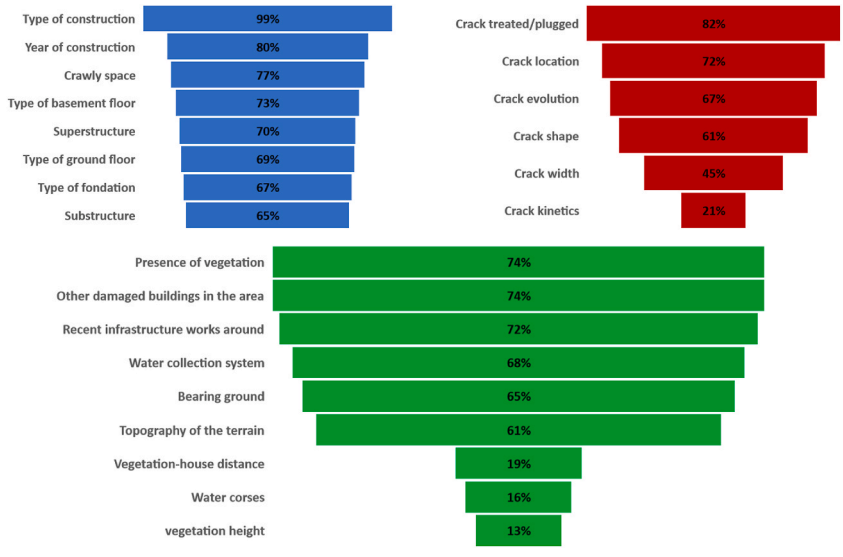


Fig. 13. Completeness overview of key data categories: Building characteristics (Left), Damage-related criteria (Right), and environmental attributes (Bottom center).

5.1. Data completeness and information gaps

The degree of data completeness in loss adjustment reports varies significantly, depending on the expert conducting the assessment and the cooperation of the insured party in providing sufficient information about the damaged property [32]. In addition certain elements, particularly those related to building foundations or subsurface parts of the building, are often inaccessible during standard visual inspection [29].

Overall, the database shows a relatively high level of completeness for building-related variables, averaging 71 %. The most consistently reported attribute is the building type (for example, detached or not), which is available in 99 % of cases. In contrast, damage-related criteria have a moderate completeness rate of 58 %, with crack kinetics being the least documented (21 %). Environmental context data is the most incomplete category, with only 47 % coverage. Notably, vegetation, often cited as a key damageability factor, is frequently missing, highlighting a critical gap in the dataset (Fig. 13).

These limitations suggest a need to supplement expert assessments with open-access data and geospatial techniques to fill in missing information and improve the analysis of clay shrink-swell phenomenon [27]. As Douglas [29] suggests, when direct observation is not feasible, such as for foundation depth, proxy indicators like building age or number of storeys may be used to infer relevant structural characteristics.

5.2. Comparison between the damage grades and the claims indemnity cost

This analysis aimed to validate the reliability of the assigned damage grades by comparing them with the **average indemnity cost per claim**. Indemnity amounts were provided by Generali Insurance Company and drawn from their claims database. To ensure consistency, all values were adjusted to **2024 euros**.

We matched indemnity records for clay shrink-swell with the damage grades derived from loss adjustment reports. Out of 10,325 cases, a subset of 6639 cases was successfully linked. The remaining cases could not be matched due to discrepancies between the two databases.

The relationship between damage grades and indemnity costs was analysed by plotting average claim amounts against damage levels (Fig. 14). As expected, results show a clear positive correlation: higher damage grades correspond to higher average indemnity costs. This trend supports the internal consistency of our damage classification system. The correlation coefficient ($R^2 = 0.89$) confirms a strong linear association between damage severity and financial loss.

It is important to clarify that the “D0” category does not indicate the absence of damage. Rather, it results from the weighted averaging method used to synthesize multiple damage criteria (as described in Section 4.4). In some cases, the computed average was close to zero, leading to a D0 classification. These records, along with those labelled as “No information,” were excluded from further analysis but are shown here to illustrate the full distribution of the matched dataset (see Table 4). The final sample of 6639 cases forms the analytical foundation for the statistical assessments presented in the subsequent sections. This dataset is sufficiently robust for the objectives of this study.

5.3. Linking damageability parameters to observed damages

This section explores the relationship between damage grades and specific building characteristics and environmental context to identify key damageability factors related to clay shrink-swell. Using the 6639 matched cases (as introduced in Section 5.2), we analysed how structural variables, such as construction period, number of storeys, foundation type, and wall materials, correlate with observed damage levels.

Each parameter was examined individually against the assigned damage grades. While the results reveal meaningful associations, it is important to note that these findings represent correlations, not causal relationships. Establishing causality will be the focus of future work.

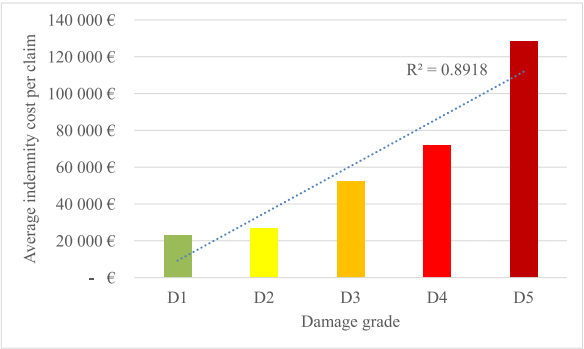


Fig. 14. Correlation between the average indemnity cost per claim and the degrees of damage.

Table 4
Number of claims per damage levels and their relative values calculated excluding the “no-information”.

Damage grade	Number of claims	Number of claims (%)
D0	618	11
D1	1181	20
D2	2532	43
D3	906	15
D4	618	11
D5	12	0
No information	772	0

5.3.1. Year of construction

Buildings in the database were classified into seven construction period categories: **prior to 1920**, **between 1920 and 1945**, **between 1945 and 1975**, **after 1975**, **after 2008**, **mixed** (indicating multiple years), and **no information**. The first five categories were derived from expert inputs in loss adjustment reports. The “after 2008” and “mixed” categories were added based on expert recommendations and literature to better reflect recent construction trends. Although the “after 2008” category is technically a subset of “after 1975”, the available data did not allow for a more granular separation between 1975 and 2008. Given the importance of analysing recent buildings, this specific category was included when the year of construction was explicitly indicated.

Excluding the records with "no information" category, most claims are associated with buildings constructed "after 1975" (2877 claims,52 %). This is followed by the "prior to 1920" (1109 claims, 20 %), "between 1945 and 1975" (1090 claims, 20 %), "between 1920 and 1945" (195 claims,3.4 %), "mixed" includes 150 claims (2.6 %), and "after 2008″ has 131 claims (2 %).

Analysis of damage grade distributions by construction period reveals several key patterns. The “after 2008” category shows a clear concentration in D1 (slight damage), suggesting improved structural resistance in newer buildings. Across all categories, D2 (moderate damage) is the most common grade, with proportions ranging from 30 % to 42 %, indicating a general vulnerability trend at this level. Older buildings, particularly those built before 1920, show a decreasing trend in claim frequency from D1 to D4, with a slight increase at D2. Categories such as “between 1920 and 1945”, “between 1945 and 1975”, and “after 1975” show relatively uniform distributions across all damage grades, indicating a wide variability in structural performance regardless of construction period (see 5Fig. 15).

These patterns highlight the relevance of construction age as a damageability factor, with newer buildings showing signs of better performance, although still susceptible to light or moderate levels of shrink-swell related damage.

5.3.2. Foundation types

Our database documents eight categories of foundation types: reinforced strip footing, unreinforced strip footing, heterogeneous in stones and/or rubble, mixed, beams, raft foundation, micropiles, and no information. The claims frequency for each foundation type is summarized in Table 5. Excluding the "no information" category, most claims were recorded in the "reinforced strip footing" category, representing 52 % of all claims. When considering both reinforced and unreinforced strip footings together, this category accounts for 60 % of the claims. The "heterogeneous in stones and/or rubble" category had the second-highest number of claims, accounting for 17 %. Only four cases were recorded for the "micropile" foundation type.

The correlation between damage grades and foundation types reveals distinct patterns. Heterogeneous foundations, composed of stones and rubble, show a uniform distribution of damage across all levels. Both reinforced and unreinforced strip footings also display an even distribution across grades D1 to D4. Conversely, composite (mixed) foundations concentrate damage at grades D2, D3, and D4, indicating lower effectiveness. Raft foundations are primarily found in grade D2, with similar frequencies in grades D1 and D3.

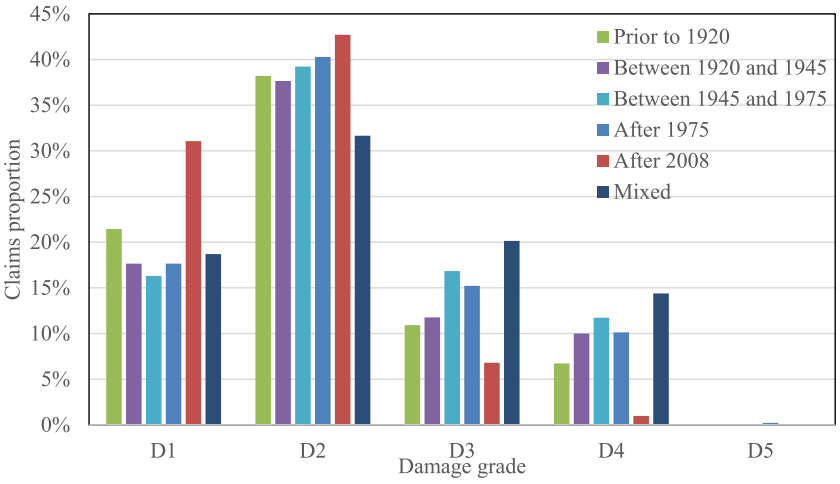


Fig. 15. Damage grade distribution by year of construction.

Table 5
Claims frequency per foundation types.

Foundation types	Claims frequency
Reinforced strip footing	3456
Unreinforced strip footing	501
Heterogeneous in stones and/or rubble	1110
Mixed	245
Beams	100
Raft foundation	50
Micropiles	4
No information	1173

Micropiles perform best, predominantly appearing in grades D1 and D2 (Fig. 16).

5.3.3. *Foundation depth*

Our results confirmed that foundation depth significantly influences building vulnerability to clay shrinkage and swelling. Buildings with varying foundation depths exhibited higher damageability than those with uniform depths (Fig. 17). At the slight damage grade D1, approximately 17 % of buildings had a single foundation depth, while nearly 12 % had two different depths. As damage severity increased, the proportion of buildings with a single depth decreased, whereas those with multiple depths increased. For instance, at grade D5, single-depth foundations accounted for 12 %, while multiple-depth foundations accounted for 19 %.

To refine this analysis, we further categorized foundation depths into three classes: $\leq$1.5 m, between 1.5 and 3 m, and >3 m, using only the subset of buildings for which depth data were available (Fig. 18). Due to the limited sample size for D5, we focused the analysis up to grade D4. The results show that buildings with foundations deeper than 3 m experienced fewer D4 cases compared to shallower depths. Notably, foundations between 1.5 and 3 m showed more damage than those >3 m, but both categories presented less damage overall than foundations $\leq$1.5 m. Interestingly, the between 1.5 and 3 m group displayed the highest proportion of slight damage (D1), nearly twice that of the >3 m group and three times that of the $\leq$1.5 m group, suggesting that moderate-depth foundations may initially absorb shrink-swell effects but remain vulnerable to long-term differential movement.

5.3.4. *Infrastructures*

Infrastructures, including building substructure systems below ground level, are made from various materials with differing strengths against clay shrinkage and swelling. Most reported claims involved "hollow concrete blocks" (57 %) and "stone and rubble" (28 %). The frequency of claims by substructure material is summarized in Table 6 below.

Damage grades across infrastructure types confirm that, for both subsurface materials (hollow concrete blocks and stone and rubble), damages can occur at all levels from D1 to D4, with a concentration around grade D2 (Fig. 19). Interestingly, the results show no significant difference in damage severity between the two materials, despite their nearly equal representation in the dataset. This challenges the common perception that stone and rubble structures are more vulnerable. The findings suggest that their lower rigidity and typically greater thickness may contribute to a damage profile comparable to that of hollow concrete blocks.

5.3.5. *Sanitary vacuum in the building (crawly space)*

Contrary to expectations, there is no huge difference in damage levels between buildings with a sanitary vacuum and those without. The distributions are almost similar across different degrees of damage, with slight increase of number of degrees of damages D3 and D4 in presence of the sanitary vacuum (Fig. 20). Although sanitary vacuums are perceived as one of the better adaptations for

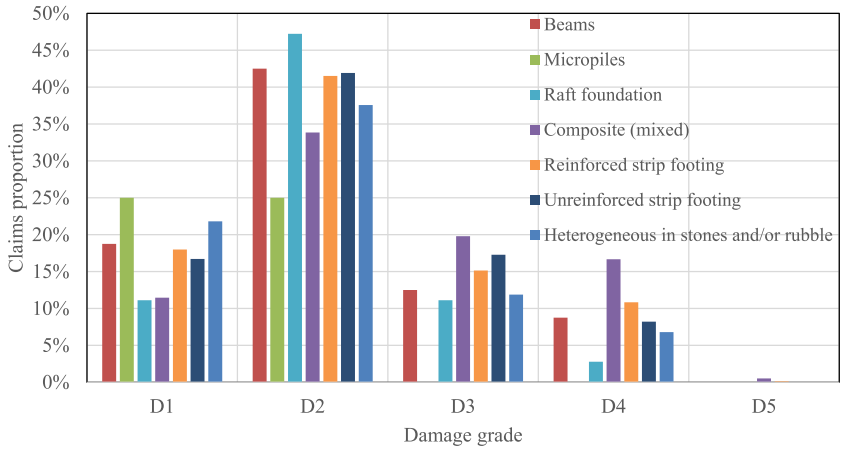


Fig. 16. Damage grade distribution by foundation types.

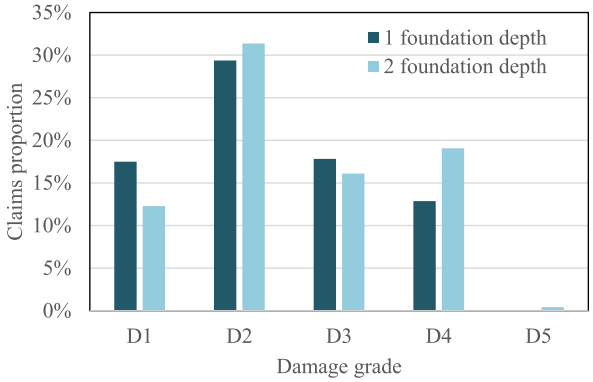


Fig. 17. Damage grade distribution by foundation depth levels.

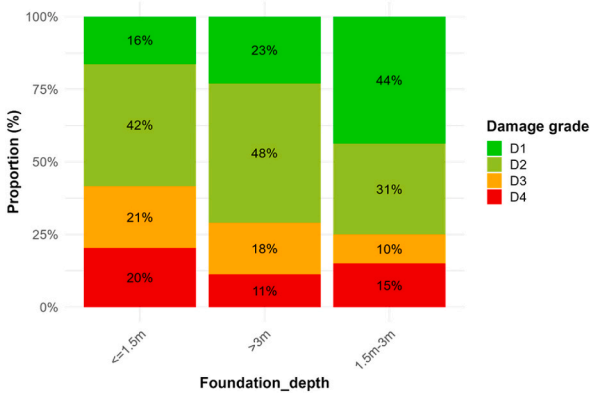


Fig. 18. Damage grade distribution by foundation depth.

Table 6
Claims distribution by foundation infrastructures materials.

Infrastructures (substructures) materials	Claims frequency
Hollow concrete blocks	2832
Stone and rubble	1377
Heterogeneous concrete: reinforced or unreinforced	298
Formwork concrete	320
Solid elements	111
Beams	4

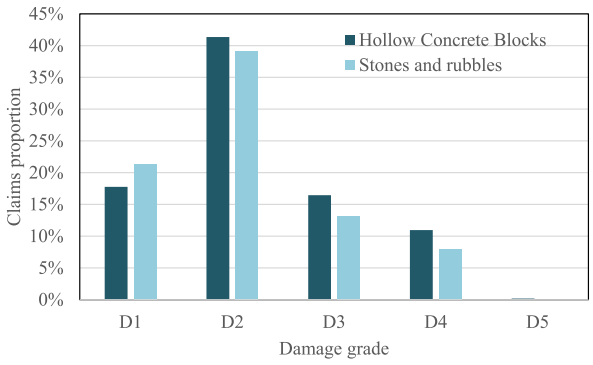


Fig. 19. Damage grade distribution by infrastructure systems.

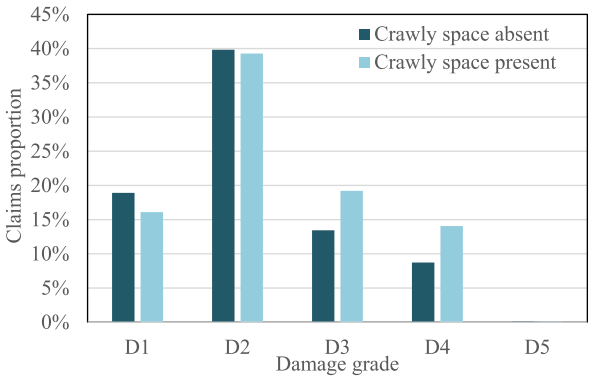


Fig. 20. Damage grade distribution by the crawly space.

mitigating foundation damage due to clay shrink-swell phenomenon, they can sometimes act as water retention areas. This disturbance of the water equilibrium underneath the foundation can exacerbate damage from this phenomenon, rather than making the building more resistant [41].

5.3.6. Extension

The correlation between damage grades and the presence of house extensions reveals a significant increase in vulnerability. Houses with extensions tend to exhibit a higher probability of sustaining more severe damage, ranging from slight to major levels (Fig. 21). It is important to clarify that the term "extension" encompasses various situations. These may include buildings with multiple declared construction years, differing foundation depths, or presence of adjoining structures such as garages or annexes.

5.3.7. Types of construction (Isolated or adjoining)

Isolated (non-adjoining) buildings are the most vulnerable to damage. As the level of damage increases, the number of damaged detached buildings also rises. In contrast, for non-detached buildings, the number of damaged buildings decreases as the damage level increases (Fig. 22).

5.3.8. Superstructures

The distribution of claims based on superstructure materials is as follows: Reinforced and unreinforced load-bearing masonry (3658 claims) and Unreinforced load-bearing masonry with rubble stone (896 claims). These materials account for nearly 63 % of all claims documented in the database. The analysis of damage grades versus superstructures indicates that the probability of experiencing damage across all levels is almost uniform for the mentioned materials (Fig. 23). Therefore, the conclusion is that the superstructure does not affect vulnerability if the foundations are sufficiently deep or well-constructed.

6. Discussion

This study leveraged empirical building damage data from Generali Insurance to develop a Clay Shrink-Swell Damage Severity Scale and to construct a detailed database of structural and environmental characteristics. Key damageability factors related to clay shrink-swell phenomenon were identified through this database.

Using Large Language Models (LLMs), we efficiently extracted relevant information from unstructured sources, improving the

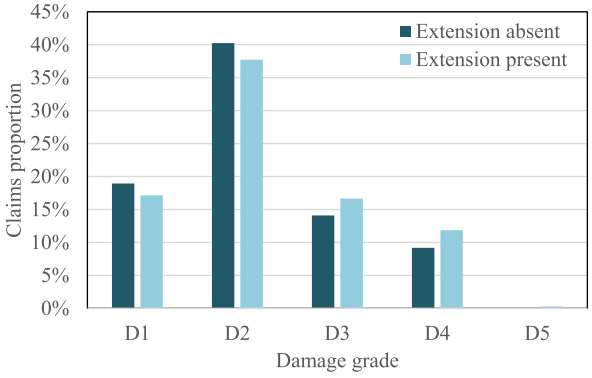


Fig. 21. Damage grade distribution by building extension.

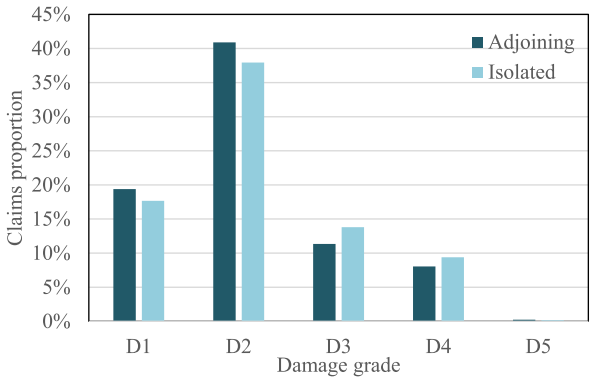


Fig. 22. Damage grade distribution by building types (detached and non-detached).

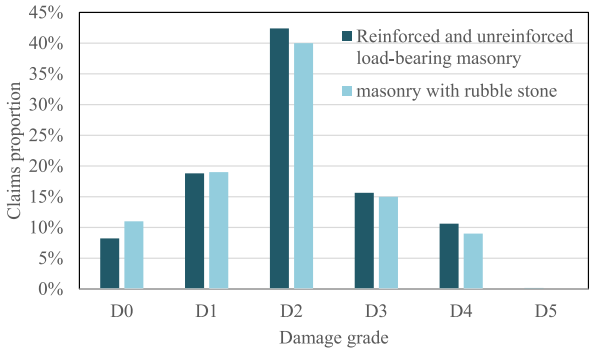


Fig. 23. Damage grade distribution by superstructure systems.

granularity and reliability of the damageability analysis.

One of the main outcomes was the development of a standardized damage severity scale, grounded in real-world observations across mainland France. This scale offers a consistent framework for assessing structural damage levels, facilitating comparability and decision-making in both research and insurance contexts.

Our findings also establish a strong association between specific building characteristics and observed damage severity. These results contribute to a better understanding of the drivers of damageability, refining prior assumptions and offering evidence-based insights into clay shrink-swell phenomenon.

Before discussing the results further, it is essential to acknowledge the study's limitations. Uncertainties stem from both the underlying data and the methodological approach. However, a formal quantification of these uncertainties lies beyond the scope of this work.

Data limitations and uncertainties include:

- **Unstructured Data Format**: The original reports were available only in Word and PDF formats, making large-scale data extraction labour-intensive and technically demanding. Despite expert intervention, it was not feasible to retrieve all relevant information from the 10,325 documents. However, key variables were identified based on expert judgment and bibliographic references to guide the extraction process.
- **Data Inconsistency**: Variability in terminology, reporting styles, and documentation detail led to inconsistencies across reports. Standardization and harmonization were necessary to ensure comparability, particularly as similar concepts were described using different vocabularies or conversely, the same terms were used with differing meanings.
- **Incomplete Data Fields**: Not all database entries were uniformly populated. Variations in data completeness were influenced by the expertise of the assessor, the property owner's knowledge, and the degree of cooperation during site visits. Information on elements such as foundation type or subsurface infrastructure was often missing.
- **Single Source Bias**: The data were sourced solely from Generali Insurance, limiting the ability to estimate the proportion of undamaged structures. While this restricts representativeness at the population level, the quality and granularity of the raw, non-aggregated data remain a valuable asset for in-depth vulnerability analysis.
- **Environmental Context**: Environmental data, such as vegetation presence, height, and proximity to buildings, are affected by temporal change. Given that the reports span over two decades, land cover dynamics may impact interpretations. Although the integration of satellite imagery could improve accuracy, the use of historical records still holds predictive value.

- **Duplicate Reports**: In cases of high-cost claims, multiple expert assessments were available for the same structure. If not properly addressed, duplicates could bias the analysis. This was mitigated by georeferencing reports and comparing associated data to retain only a single entry per case.

Methods limitations and uncertainties include:

- **Manual Reference Dataset**: The reference dataset, built from 155 manually reviewed reports, may not fully represent the diversity of the 10,325 reports in the full dataset. Damage descriptions varied considerably across experts, creating challenges in standardizing damage extraction. Although bibliographic sources and expert judgment guided the selection of key criteria, some nuances may have been overlooked.
- **Damage Severity Scale Proposition**: The proposed damage scale was derived from the 155 reference cases and extended to the entire dataset. This extrapolation could lead to over- or underestimation of damage severity in some cases. Nonetheless, the scale was anchored in both expert knowledge and existing literature to ensure its relevance and usability.
- **Prompt Engineering**: Designing effective extraction prompts required iterative refinement and significant manual effort. Despite the potential of LLMs and RAG systems, suboptimal prompt tuning introduces risk of missed or misclassified information. A trial-and-error approach was necessary to capture essential features reliably.
- **Final Damage Grades:** Final damage levels were computed by averaging four key crack-related criteria: width, kinetics, shape, and location (see Section 4.4). Missing values and the assumption of equal weighting across criteria introduced uncertainty. Grades were rounded to the nearest whole number (D1 to D5), with intermediate values near zero forming a D0 class. These D0 cases, artifacts of the averaging method, were excluded from correlation analyses as they do not reflect an absence of damage.
- **Validation of Damage Grades**: Currently, there is no clear hazard parameter for clay shrinkage and swelling in terms of ground movement, especially on a large scale. The distribution of clayey soils across the territory is used, along with meteorological factors such as the Soil Wetness Index (SWI). In this study, the physical damage from the proposed damage scale was validated using the average indemnity cost per claim. This could introduce uncertainties because the value of the buildings was not considered. For instance, minor damage to a high-value property may cost more than extensive damage to a low-value structure. Additionally, garages and annexes were treated as principal buildings, though their repair costs differ substantially.
- **Workshops with Experts:** Expert consensus was used to refine damage classification, but subjective differences persist. Some experts might rate the same damage as D1, others as D2. To minimize overestimation, we systematically applied the lower rating, aligning with the European Macroseismic Scale [44], which considers most shrink–swell damage as minor.

6.1. Creation of the database

To our knowledge, this study presents the first comprehensive and structured database dedicated to damages caused by the clay shrink–swell phenomenon in low-rise buildings across mainland France. It compiles detailed information on building characteristics, surrounding environmental conditions, and observed damage. The database provides a robust foundation for ex-post damage analysis and physical vulnerability assessment, supporting various applications such as insurance compensation, damage accounting, risk modelling, and the development of prevention and mitigation strategies for insurers, homeowners, and public authorities.

The uniqueness of this database lies in its use of insurance loss adjustment reports. These reports contain valuable historical data based on field assessments conducted by experts, yet they remain largely underutilized. Confidentiality concerns, business restrictions, and the unstructured nature of the documents (PDF and Word formats) limit access and usability. Moreover, the variability in content and terminology from one expert to another adds complexity to the extraction process. As a result, transforming these reports into structured data is both time-consuming and technically demanding.

To address these challenges, we applied recent advances in artificial intelligence, specifically Retrieval-Augmented Generation (RAG) and Large Language Models (LLMs), to support the extraction, classification, and contextual interpretation of the information. Unlike traditional text mining techniques that focus on keyword detection, our approach enabled a deeper understanding of the content, more closely resembling expert interpretation. AI was used to enhance the process, while expert knowledge remained central to the analysis.

The spatial and temporal coverage of the database ensures its national representativeness. Claim distribution corresponds closely with areas of known exposure to clay shrink–swell hazards, as shown in the 2019 exposure map by the French Geological Survey (BRGM). It also spans major drought events, including those of 2003, 2011, and 2018 [13], which are known to amplify damage caused by this phenomenon.

In addition to serving as a research asset, the database provides a standardized tool for field professionals, enabling consistent data collection through harmonized vocabulary and predefined fields. The methodology can be applied to existing reports, allowing for efficient data transformation and improved use of archival information. Looking ahead, integrating digital tools such as tablets during field assessments would facilitate direct data entry into structured formats. This would improve workflow efficiency and minimize information loss, making damage data more accessible and applicable for statistical and risk assessment studies.

6.2. Proposition of the damage scale for clay shrinkage and swelling and attribution of the damage grades

In parallel with the creation of the database, we developed a damage severity scale specifically tailored to the clay shrink-swell

phenomenon, derived from insurance loss adjustment reports. To our knowledge, this is the first time such a scale has been established using insurance data. The scale focuses exclusively on physical damage and categorizes severity into five distinct levels, from slight (D1) to severe damage (D5). One of its main advantages is the removal of economic bias, as it is based on observed physical impacts rather than financial loss.

Although the magnitude of ground movement due to clay shrink-swell processes is generally lower than that caused by phenomena such as mining subsidence [47], the associated repair and compensation costs are nonetheless considerable for insurance providers. The scale developed in this study is practical and accessible to both professionals and non-specialists. Its clarity and object-specific structure minimize confusion and facilitate consistent evaluations. By structuring damage assessment across various building components from exteriors to foundations, it enhances the readability and usability of the information. While it is often difficult to observe or record damages affecting foundations and subsurface structures [47], the scale compensates for this by incorporating four key diagnostic criteria: crack location, width, kinetics, and shape.

The national coverage of the dataset, both spatially and temporally (claims from 2000 to 2021), reinforces the relevance of the proposed scale for assessing past and future damages across mainland France. Its application beyond the French context, however, may be limited by regional differences in construction practices, geological conditions, and climatic patterns.

Importantly, the scale introduces standardized terminology and structure to damage descriptions, addressing the inconsistencies previously found in expert reports. It includes four dedicated fields for structured assessment, along with an additional field that harmonizes damage-related keywords across reports. This approach brings uniformity and clarity, reflecting real-world cases and field conditions.

In terms of observed frequencies, the most common damage level is D2, representing 43 % of all claims, followed by D1 (20 %) and D3 (15 %). Together, D0, D1, D2, and D3 account for 89 % of the dataset, with typical maximum crack widths of up to 15 mm. Only 11 % of the cases fall into the more severe categories (D4 and D5). Given this distribution, future efforts should focus on refining the classification of the most frequently occurring damage grades rather than emphasizing the less common severe cases.

6.3. Identification of damageability factors

The results from the correlation analysis between damage grades and building characteristics should be considered preliminary. They are based on bivariate analysis, which explores the relationship between two variables at a time. While this method cannot capture complex interactions or multivariate dependencies, it offers valuable initial insights into damage probabilities and influencing factors.

Key findings from this analysis include the following:

- **Year of Construction**: This parameter emerged as particularly important, serving as a proxy for other, less easily observable factors such as foundation type and construction materials. Our analysis shows that buildings constructed after 1975 constitute the majority of the sample (52 %) and exhibit claims across all damage grades. Buildings constructed after 2008 are associated predominantly with slight damage (D1), making them the least vulnerable group. Conversely, buildings constructed before 1920 also show relatively low vulnerability, with most claims in the D1 category and decreasing frequency at higher damage grades. These results are consistent with existing literature: Crilly (2001) suggested that pre-1900 buildings are less vulnerable due to the flexibility of the materials used in traditional mortar. Additionally, recent building code updates (e.g., DTU 20.1), which mandate vertical reinforcement at ground-floor corners since 2008, may also contribute to reduced vulnerability [41].
- **Building Extensions**: Extensions were found to increase building vulnerability, likely due to structural discontinuities that act as weak points during ground movement. In our study, the “mixed” category used when multiple values were recorded for a single variable, often signified the presence of an extension. For instance, multiple construction years or foundation depths within a single case suggest that an addition was made to the original structure, potentially compromising its overall stability.
- **Construction Materials and Substructure Characteristics**: The type of infrastructure and superstructure materials, along with the presence or absence of a sanitary void (vide sanitaire), showed relatively even distributions across damage grades. This suggests that, based solely on bivariate analysis, damage can occur at any level regardless of these parameters. It was therefore not possible to determine which specific material or configuration was more vulnerable within the scope of this preliminary approach.

7. Conclusions

This study presents the first comprehensive database and physical damage scale dedicated to assessing building damageability associated with the clay shrink–swell phenomenon in mainland France. Based on insurance loss adjustment reports, the database integrates both structural and environmental characteristics, providing a solid empirical foundation for damage analysis. The proposed damage scale, developed from real-world cases and expert observations, offers a standardized reference for assessing damage severity and supports more consistent and systematic data collection by loss adjusters. Despite variations in the level of detail across reports, the scale contributes to harmonizing damage descriptions and terminology.

The application of Large Language Models (LLMs) significantly enhanced the extraction and structuring of unstructured textual data. These models effectively captured contextual meaning and linguistic patterns, although their performance relied on expert supervision, careful prompt design, and iterative refinement. This demonstrates the complementary roles of artificial intelligence and domain expertise in transforming raw documents into structured, meaningful datasets.

The correlation between damage grades and indemnity costs confirms the practical value of the proposed scale, with higher grades

consistently linked to increased compensation. Preliminary analyses also revealed notable trends in damageability based on construction year, building materials, and the presence of extensions or adjoining structures. However, a more complete understanding of the complex interactions between these factors will require multivariate analysis. Such approaches will improve predictive accuracy and support better-informed mitigation strategies. It is important to note that correlation does not imply causation, and further investigation is needed to establish causal relationships.

Future research could be strengthened by incorporating geotechnical investigation data to complement the findings drawn from insurance reports. Information such as foundation depths, root penetration, typically available from site-specific studies, can offer greater detail on environmental and structural conditions. However, these data are often inaccessible at large scales and require significant time and financial resources to obtain. These limitations highlight why such information is rarely included in large-scale studies like this one. Nonetheless, whenever available, their use could significantly refine the analysis and improve the identification of key damageability factors.

Finally, it is important to emphasize that the clay shrink-swell phenomenon is projected to become more frequent and intense under climate change. As extreme weather events, including prolonged droughts and high variability in soil moisture, are expected to increase, so too will the associated structural risks. This reinforces the long-term relevance of the present work and highlights the urgent need to improve damage assessment tools and risk-informed planning strategies in a changing climate.

CRediT authorship contribution statement

Noel Philmon Mzungu: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Caterina Negulescu:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Etienne Bertrand:** Writing – review & editing, Supervision, Methodology, Formal analysis, Data curation, Conceptualization. **Arthur Perringaux:** Visualization, Methodology, Formal analysis, Data curation, Conceptualization. **Laurent Boissier:** Writing – review & editing, Supervision, Resources, Data curation, Conceptualization. **Sébastien Gourdier:** Writing – review & editing, Resources, Methodology. **Bastien Colas:** Writing – review & editing, Resources, Methodology, Conceptualization. **Frédéric Leone:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization. **Gilles Grandjean:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

This research is part of the CIFRE-PhD thesis of Noel Philmon Mzungu, co-funded by Generali France, BRGM (Bureau de Recherches Géologiques et Minières), and LAGAM (Laboratoire de Géographie et Aménagement de Montpellier). The author would like to thank these institutions for providing access to their data. Additionally, the author thanks Dr. Shihab Sylvain for his assistance with Generative AI technology, and Quentin Henaff for insightful discussions on geography and technical help in exploring historical climate claims data.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijdrr.2025.105636.

Data availability

Data will be made available on request.

References

[1] D.G. Toll, Z. Abedin, J. Buma, Y. Cui, A.S. Osman, K.K. Phoon, The impact of changes in the water table and soil moisture on structural stability of buildings and foundation systems: systematic review CEE10-005 (SR90). https://durham-repository.worktribe.com/output/1636508, 2012.
[2] M. Duc, *Les argiles dans le génie civil: Pathologies et propriétés remarquables* [Thesis, Université Paris Est - Marne-la-Vallée]. https://theses.hal.science/tel-03089797, 2020.
[3] J.B. Burland, Behaviour of Foundations and Structures, 1977.
[4] W.A. Charlie, M.A. Osman, E.M. Ali, Construction on expansive soils in Sudan, J. Construct. Eng. Manag. 110 (3) (1984) 359–374, https://doi.org/10.1061/(ASCE)0733-9364(1984)110:3(359.
[5] A. Das, S. Roy, Effect of Expansive Soil on Foundation and its Remedies, vol. 6, 2014.
[6] J.C. Doornkamp, Clay shrinkage induced subsidence, Geogr. J. 159 (2) (1993) 196–202, https://doi.org/10.2307/3451410.

[7] K. Faye, F. Samb, P.S. Faye, Y. Berthaud, Factors predisposing and triggering the phenomenon of shrinkage-swelling of clay soils in the urban center of diamniadio, Geomaterials 13 (2) (2023) 23–34, https://doi.org/10.4236/gm.2023.132002.
[8] B. Kalantari, Foundations on expansive soils: a review, Res. J. Appl. Sci. Eng. Technol. 4 (2012).
[9] A.N. Karunarathne, Investigation of Expansive Soil for Design of Light Residential Footings in Melbourne, 2016.
[10] N. Ravichandran, V. Mahmoudabadi, S. Shrestha, Analysis of the bearing capacity of shallow foundation in unsaturated soil using monte carlo simulation, Int. J. Geosci. 8 (10) (2017) 1231–1250, https://doi.org/10.4236/ijg.2017.810071.
[11] L. Jones, V. Banks, I. Jefferson, Chapter 8 swelling and shrinking soils, Geological Society, London, Engineering Geology Special Publications 29 (1) (2020) 223–242, https://doi.org/10.1144/EGSP29.8.
[12] L. Ighil Ameur, Analysis of climate change impacts on the shrinkage-swelling phenomenon of clayey soils to adapt infrastructures, in: 8th International Conference on Unsaturated Soils (UNSAT 2023), vol. 382, 2023, https://doi.org/10.1051/e3sconf/202338201003, 01003.
[13] J.-L. Salagnac, Lessons from the 2003 heat wave: a French perspective, Build. Res. Inf. 35 (4) (2007) 450–457, https://doi.org/10.1080/09613210601056554.
[14] T. Corti, M. Wüest, D. Bresch, S.I. Seneviratne, Drought-induced building damages from simulations at regional scale, Nat. Hazards Earth Syst. Sci. 11 (12) (2011) 3335–3342, https://doi.org/10.5194/nhess-11-3335-2011.
[15] France Assureur, Master class # 2: « La sécheresse et son impact sur les habitations: quelles solutions de prévention et de protection?. https://www.franceassureurs.fr/wp-content/uploads/2023-10-17-master-class-rga-vdef-print.pdf, 2023.
[16] BRGM, Exposition au retrait-gonflement des argiles | InfoTerre. https://infoterre.brgm.fr/actualites/exposition-au-retrait-gonflement-argiles, 2019.
[17] T. Corti, V. Muccione, P. Köllner-Heck, D. Bresch, S.I. Seneviratne, Simulating past droughts and associated building damages in France, Hydrol. Earth Syst. Sci. 13 (9) (2009) 1739–1747, https://doi.org/10.5194/hess-13-1739-2009.
[18] S. Barthélémy, B. Bonan, J.-C. Calvet, G. Grandjean, D. Moncoulon, D. Kapsambelis, S. Bernardie, A new drought index fitted to clay shrinkage induced subsidence over France: benefits of interactive leaf area index. https://doi.org/10.5194/egusphere-2023-1366, 2023.
[19] A. Charpentier, M. James, H. Ali, Predicting drought and subsidence risks in France, Nat. Hazards Earth Syst. Sci. 22 (7) (2022) 2401–2418, https://doi.org/10.5194/nhess-22-2401-2022.
[20] Q.V. Do, *Impacts des mouvements de terrains sur une structure type 'maison individuelle': modélisation de l'interaction sol-structure pour l'évaluation de la vulnérabilité du bâti* [These de doctorat, Paris Est]. https://theses.fr/2011PEST1075, 2011.
[21] C. Negulescu, E. Foerster, Parametric studies and quantitative assessment of the vulnerability of a RC frame building exposed to differential settlements, Nat. Hazards Earth Syst. Sci. 10 (9) (2010) 1781–1792, https://doi.org/10.5194/nhess-10-1781-2010.
[22] R. Agliata, A. Bortone, L. Mollo, Indicator-based approach for the assessment of intrinsic physical vulnerability of the built environment to hydro-meteorological hazards: review of indicators and example of parameters selection for a sample area, Int. J. Disaster Risk Reduct. 58 (2021) 102199, https://doi.org/10.1016/j.ijdrr.2021.102199.
[23] S. Bera, B. Guru, T. Oommen, Indicator-based approach for assigning physical vulnerability of the houses to landslide hazard in the Himalayan region of India, Int. J. Disaster Risk Reduct. 50 (2020) 101891, https://doi.org/10.1016/j.ijdrr.2020.101891.
[24] S. Ettinger, L. Mounaud, C. Magill, A.-F. Yao-Lafourcade, J.-C. Thouret, V. Manville, C. Negulescu, G. Zuccaro, D. De Gregorio, S. Nardone, J.A.L. Uchuchoque, A. Arguedas, L. Macedo, N. Manrique Llerena, Building vulnerability to hydro-geomorphic hazards: estimating damage probability from qualitative vulnerability assessment using logistic regression, J. Hydrol. 541 (2016) 563–581, https://doi.org/10.1016/j.jhydrol.2015.04.017.
[25] S. Fuchs, M. Keiler, R. Ortlepp, R. Schinke, M. Papathoma-Köhle, Recent advances in vulnerability assessment for the built environment exposed to torrential hazards: challenges and the way forward, J. Hydrol. 575 (2019) 587–595, https://doi.org/10.1016/j.jhydrol.2019.05.067.
[26] C. Guillard-Gonçalves, J.L. Zêzere, S. Pereira, R.A.C. Garcia, Assessment of physical vulnerability of buildings and analysis of landslide risk at the municipal scale: application to the Loures municipality, Portugal, Nat. Hazards Earth Syst. Sci. 16 (2) (2016) 311–331, https://doi.org/10.5194/nhess-16-311-2016.
[27] F. Leone, F. Lavigne, R. Paris, J.-C. Denain, F. Vinet, A spatial analysis of the December 26th, 2004 tsunami-induced damages: lessons learned for a better risk assessment integrating buildings vulnerability, Appl. Geogr. 31 (1) (2011) 363–375, https://doi.org/10.1016/j.apgeog.2010.07.009.
[28] M. Uzielli, F. Nadim, S. Lacasse, A.M. Kaynia, A conceptual framework for quantitative estimation of physical vulnerability to landslides, Eng. Geol. 102 (3–4) (2008) 251–256, https://doi.org/10.1016/j.enggeo.2008.03.011.
[29] J. Douglas, Physical vulnerability modelling in natural hazard risk assessment, Nat. Hazards Earth Syst. Sci. 7 (2) (2007) 283–288, https://doi.org/10.5194/nhess-7-283-2007.
[30] M.B. Malgwi, S. Fuchs, M. Keiler, A generic physical vulnerability model for floods: review and concept for data-scarce regions, Nat. Hazards Earth Syst. Sci. 20 (7) (2020) 2067–2090, https://doi.org/10.5194/nhess-20-2067-2020.
[31] B. Merz, H. Kreibich, R. Schwarze, A. Thieken, Review article 'Assessment of economic flood damage', Nat. Hazards Earth Syst. Sci. 10 (8) (2010) 1697–1724, https://doi.org/10.5194/nhess-10-1697-2010.
[32] C. André, D. Monfort, M. Bouzit, C. Vinchon, Contribution of insurance data to cost assessment of coastal flood damage to residential buildings: insights gained from Johanna (2008) and Xynthia (2010) storm events, Nat. Hazards Earth Syst. Sci. 13 (8) (2013) 2003–2012, https://doi.org/10.5194/nhess-13-2003-2013.
[33] A. Heranval, *Contributions des données de l'assurance à l'étude des risques naturels: application de méthodes d'apprentissage statistique pour l'évaluation de la nature et du coût des dommages assurés liés aux événements naturels en France* [Phdthesis, Sorbonne Université]. https://theses.hal.science/tel-03935722, 2022.
[34] D. Zappa, M. Borrelli, G.P. Clemente, S. Nino, Text Mining in Insurance: from Unstructured Data to Meaning, 2019.
[35] M. M. de Brito, C. Kuhlicke, A. Marx, Near-real-time drought impact assessment: a text mining approach on the 2018/19 drought in Germany, Environ. Res. Lett. 15 (10) (2020) 1040a9, https://doi.org/10.1088/1748-9326/aba4ca.
[36] M. M. de Brito, J. Sodoge, H. Kreibich, C. Kuhlicke, *Text-mining of natural hazard impacts (TM-Impacts): an application to the 2021 flood in Germany* (EGU22-2001). EGU22, Copernicus Meetings (2022), https://doi.org/10.5194/egusphere-egu22-2001.
[37] C. Balona, ActuaryGPT: applications of large language models to insurance and actuarial work, Br. Actuar. J. 29 (2024) e15, https://doi.org/10.1017/S1357321724000102.
[38] J. Dagdelen, A. Dunn, S. Lee, N. Walker, A.S. Rosen, G. Ceder, K.A. Persson, A. Jain, Structured information extraction from scientific text with large language models, Nat. Commun. 15 (1) (2024) 1418, https://doi.org/10.1038/s41467-024-45563-x.
[39] K.K. Harrod, P. Bhandari, A. Anastasopoulos, From text to maps: LLM-driven extraction and geotagging of epidemiological data, in: D. Dementieva, O. Ignat, Z. Jin, R. Mihalcea, G. Piatti, J. Tetreault, S. Wilson, J. Zhao (Eds.), Proceedings of the Third Workshop on NLP for Positive Impact, Association for Computational Linguistics, 2024, pp. 258–270, https://doi.org/10.18653/v1/2024.nlp4pi-1.24.
[40] Y. Gao, Y. Xiong, X. Gao, K. Jia, J. Pan, Y. Bi, Y. Dai, J. Sun, M. Wang, H. Wang, Retrieval-augmented generation for large language models: a survey, arXiv: 2312.10997) (2024), https://doi.org/10.48550/arXiv.2312.10997 arXiv.
[41] MRN, Diagnostic pour L'Adaptation des habitations au retrait gonflement des argiles avant dommages. https://www.mrn.asso.fr/wp-content/uploads/2023/11/rapport-gonflement-vf.pdf, 2023.
[42] M. Gilbert, A. Crutchfield, B. Luo, K. Thind, A.I. Ghanem, F. Siddiqui, Using a Large Language Model (LLM) for automated extraction of discrete elements from clinical notes for creation of cancer databases, Int. J. Radiat. Oncol. Biol. Phys. 120 (2) (2024) e625, https://doi.org/10.1016/j.ijrobp.2024.07.1375.
[43] M.U. Hadi, Q.A. Tashi, R. Qureshi, A. Shah, A. Muneer, M. Irfan, A. Zafar, M.B. Shaikh, N. Akhtar, J. Wu, S. Mirjalili, A survey on large language models: applications, challenges, limitations, and practical usage. https://doi.org/10.36227/techrxiv.23589741.v1, 2023.
[44] G. Grünthal, European Macroseismic Scale 1998, 1998.
[45] R. Blong, A Review of Damage Intensity Scales, 2001.
[46] T.J. Freeman, R.M.C. Driscoll, G.S. Littlejohn, Has your House Got Cracks? A Homeowner's Guide to Subsidence and Heave Damage vol. 2, Telford, 2002.
[47] A.H. Cooper, The classification, recording, databasing and use of information about building damage caused by subsidence and landslides, Q. J. Eng. Geol. Hydrogeol. 41 (3) (2008) 409–424, https://doi.org/10.1144/1470-9236/07-223.
[48] B.G. Pantoja-Rosero, K.R.M. Dos Santos, R. Achanta, A. Rezaie, K. Beyer, Determining crack kinematics from imaged crack patterns, Constr. Build. Mater. 343 (2022) 128054, https://doi.org/10.1016/j.conbuildmat.2022.128054.

[49] Alheib, M., Caudron, M., & Kazmierczak, J.-B. (2018). Impact des mouvements de terrains sur les bâtis et les infrastructures. 4.
[50] H. Assadollahi, H. Nowamooz, Long-term analysis of the shrinkage and swelling of clayey soils in a climate change context by numerical modelling and field monitoring, Computers and Geotechnics 127 (2020) 103763, https://doi.org/10.1016/j.compgeo.2020.103763.
[51] M. Crilly, Analysis of a database of subsidence damage, Structural Survey 19 (1) (2001) 7–15.
[52] Emad Jahangir, O. Deck, F. Masrouri, Influence of building stiffness and foundation embedding on clays shrinkage-swelling hazard consequences, in: In: Vogt, Norbert; Schuppener, Bernd; Straub, Daniel; Bräu, Gerhardt (Hg.): Geotechnical Safety and Risk. ISGSR 2011. Karlsruhe: Bundesanstalt für Wasserbau, 2011, pp. 117–124.
[53] P. Kellermann, K. Schröter, A.H. Thieken, S.-N. Haubrock, H. Kreibich, The object-specific flood damage database HOWAS 21, Natural Hazards and Earth System Sciences 20 (9) (2020) 2503–2519. https://doi.org/10.5194/nhess-20-2503-2020.
[54] S. Plante, Subsidence case studies: using soil suction techniques, Structural Survey 16 (3) (1998) 141–145.
[55] P. Webb, Hoopsafe beams to rectify subsidence damage in low-rise buildings, Structural Survey 17 (2) (1999) 109–116.